

ERICA CHIN

SOFTWARE ENGINEERING STUDENT (42KL) | SOFTWARE ENGINEER

Kuala Lumpur, Malaysia

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PROFILE / SUMMARY

Software engineering student at 42 Kuala Lumpur building Python and C projects, including a schema-validated local LLM function-calling pipeline and a multithreaded scheduler in C. Strong in algorithms and concurrency. Brings 10+ years of marketing and data-insight work, analysing user behaviour and supporting campaigns that reached 17M+ users, to build software that serves business goals.

TECHNICAL SKILLS

- **Languages:** Python, C, JavaScript, HTML, CSS
 - **AI & Data:** LLM function calling, constrained decoding, structured output (JSON), Pydantic, data validation
 - **Systems & Algorithms:** POSIX threads, mutexes, multithreading, DFS, BFS, Prim, priority queues
 - **Tools:** Git, Linux, Make, uv, mypy, flake8, AddressSanitizer, ThreadSanitizer
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KEY PROJECTS

Call Me Maybe: LLM Function-Calling System

Tech Stack: Python, Qwen3-0.6B, Pydantic, constrained decoding

github.com/ericachinlk/42-call-me-maybe

- Built a pipeline that converts natural-language prompts into schema-validated JSON function calls using a local Qwen3-0.6B model.
- Implemented constrained decoding with token-level logit masking, using separate rules for function names, numbers and text values to reduce invalid output.
- Validated parameter types and structure with Pydantic before emitting final JSON; supports multiple local models.
- Reached about 99% accuracy on the project test set (11 prompts across 5 function types).

Codexion: Multithreaded Resource Simulation

Tech Stack: C, POSIX Threads, Mutexes, Priority Queue, AddressSanitizer, ThreadSanitizer

github.com/ericachinlk/42-codexion

- Built a multithreaded C simulation where coder threads compete for shared resources, using POSIX threads and mutexes to protect shared state.
- Designed resource acquisition so each coder takes both required resources together, preventing deadlock.
- Implemented FIFO and Earliest Deadline First (EDF) scheduling with a custom min-heap priority queue, plus timeout monitoring and thread-safe logging.
- Tested for memory errors and data races with AddressSanitizer and ThreadSanitizer.

A-maze-ing: Configurable Maze Generator (team project)

Tech Stack: Python, DFS, Prim, BFS, CLI, Python packaging

github.com/ericachinlk/42-a-maze-ing

- Built the maze generation engine, BFS pathfinding and package structure, working with a teammate who handled terminal display.
 - Delivered a configurable command-line application and reusable Python library that generate perfect and non-perfect mazes using DFS and Prim's algorithm.
 - Stored each cell's four walls in a compact 4-bit bitmask; added configuration validation and type checking (mypy).
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PROFESSIONAL EXPERIENCE

AirAsia MOVE

Assistant Manager, Marketing

2022 – 2025

- Analysed campaign and user-behaviour data to find growth opportunities and improve performance for the Hotels and Shop products across the region.
- Supported campaigns that delivered a 90%+ daily gross booking value (GBV) uplift and 70%+ booking growth compared with the previous campaign.
- Contributed to a platform campaign that reached 17M+ users in one week.

Inculab

Social Media Specialist

2021 – 2022

- Evaluated audience engagement and conversion data, increasing enquiry rate by 116% in six months and link clicks by 259% in three months.

Joey Yap Research Group

Project Lead, Content Development

2017 – 2020

- Led content projects (books, infographics, workbooks, digital products) with design and digital teams from planning to delivery; maintained a WordPress membership website.

Social Video Lab

Senior Media & Content Specialist

2016 – 2017

- Developed digital content and social campaigns for retail, cinema and publishing brands.

Oriental Daily

Senior World News Writer

2013 – 2016

- Researched, translated and verified international news from global sources for daily publication.
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EDUCATION & TRAINING

42 Kuala Lumpur

2025 – Present

- Project-based curriculum covering C, Python, algorithms, data structures and software development.
- Completed Web Discovery Piscine: Learned HTML, CSS and JavaScript; built a resume website as the final project.

Universiti Tunku Abdul Rahman (UTAR)

2010

Bachelor of Communications (Hons), Journalism
